

LAB EXPERIMENTS

Q13) Write a C program for Hill cipher succumbs to a known plaintext attack if sufficient plaintext– ciphertext pairs are provided. It is even easier to solve the Hill cipher if a chosen plaintext attack can be mounted.

Aim:

To write a C program to demonstrate a **Known Plaintext Attack** on the Hill Cipher by recovering the encryption key matrix using known plaintext and corresponding ciphertext pairs.

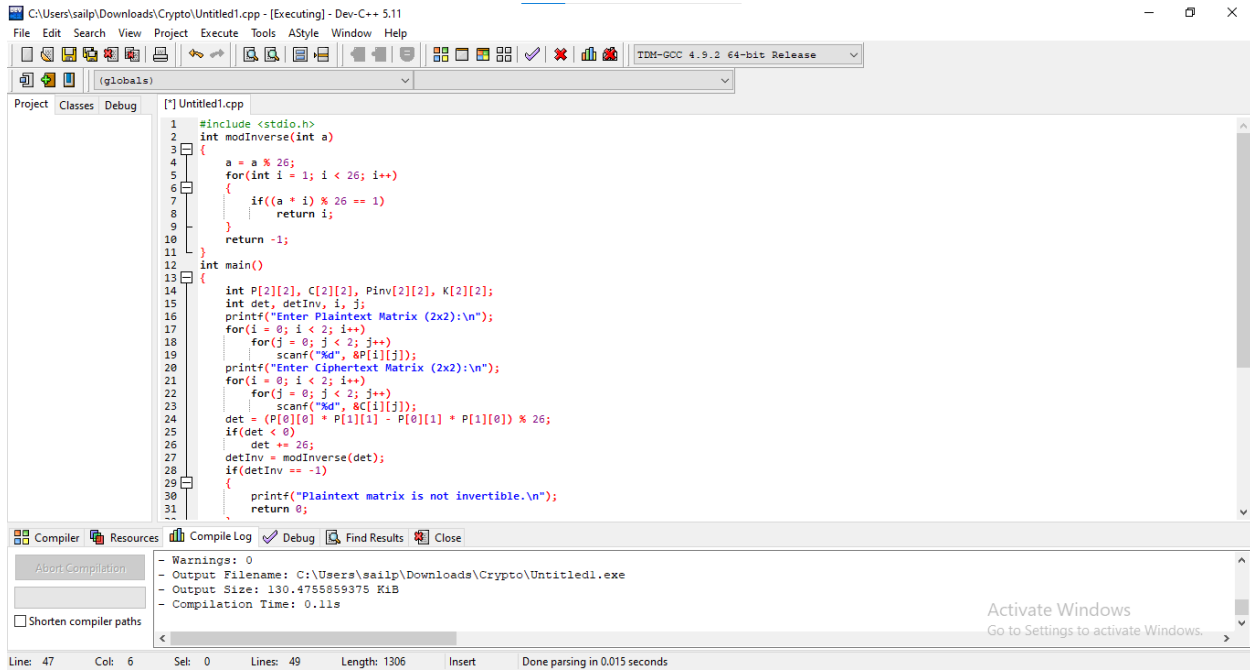
Algorithm:

1. Start the program.
2. Enter two plaintext digraphs and their corresponding ciphertext digraphs.
3. Convert each letter into its numerical equivalent ($A = 0, B = 1, \dots, Z = 25$).
4. Form the plaintext matrix (P) and ciphertext matrix (C).
5. Compute the determinant of the plaintext matrix.
6. Find the modular inverse of the determinant modulo 26.
7. Compute the inverse of the plaintext matrix (P^{-1}) modulo 26.
8. Recover the key matrix using the formula:
$$K = C \times P^{-1} \pmod{26}$$
9. Display the recovered key matrix.
10. Stop the program.

Result:

Thus, the C program to demonstrate a **Known Plaintext Attack** on the Hill Cipher was successfully executed. Using known plaintext and corresponding ciphertext pairs, the original 2×2 **key matrix** was successfully recovered.

Code:

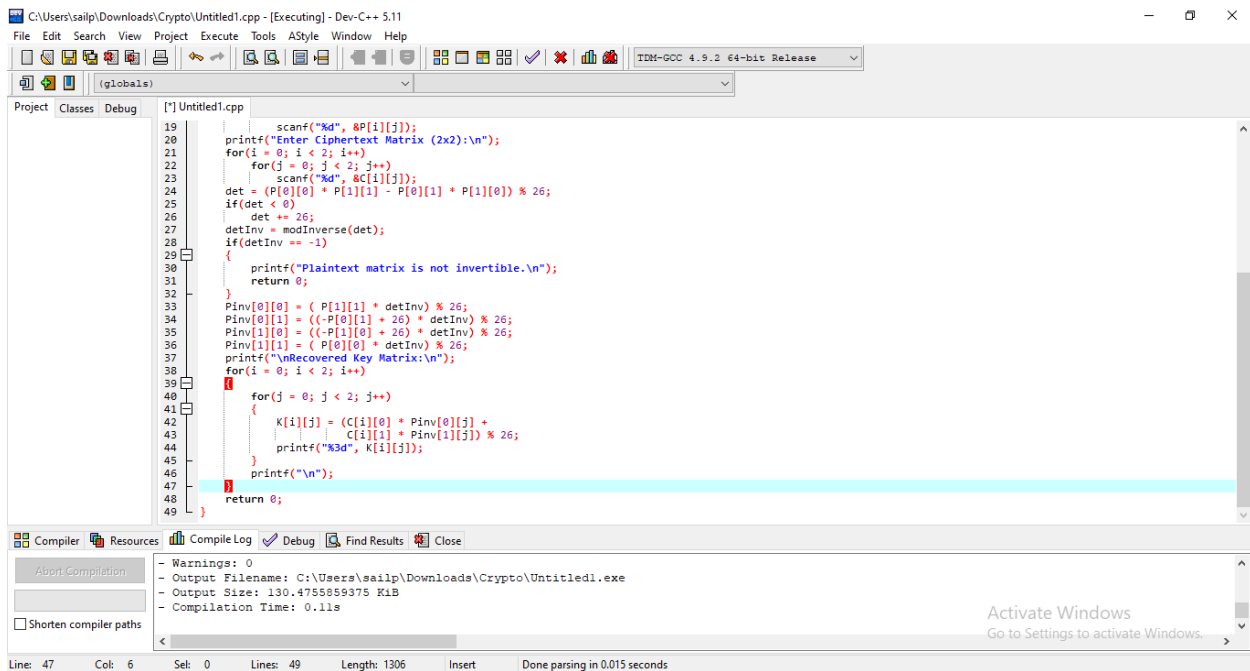


The screenshot shows the Dev-C++ IDE with the file 'Untitled1.cpp' open. The code defines a function `modInverse` and a `main` function. The `main` function prompts the user to enter a 2x2 Plaintext Matrix and a 2x2 Ciphertext Matrix. It then calculates the determinant of the ciphertext matrix and checks if it is invertible. If not, it prints an error message and returns 0. If invertible, it calculates the modular inverse of the determinant.

```
1 #include <stdio.h>
2 int modInverse(int a)
3 {
4     a = a % 26;
5     for(int i = 1; i < 26; i++)
6     {
7         if((a * i) % 26 == 1)
8             return i;
9     }
10    return -1;
11 }
12
13 int main()
14 {
15     int P[2][2], C[2][2], Pinv[2][2], K[2][2];
16     int det, detInv, i, j;
17     printf("Enter Plaintext Matrix (2x2):\n");
18     for(i = 0; i < 2; i++)
19     {
20         for(j = 0; j < 2; j++)
21             scanf("%d", &P[i][j]);
22     }
23     printf("Enter Ciphertext Matrix (2x2):\n");
24     for(i = 0; i < 2; i++)
25     {
26         for(j = 0; j < 2; j++)
27             scanf("%d", &C[i][j]);
28     }
29     det = (P[0][0] * P[1][1] - P[0][1] * P[1][0]) % 26;
30     if(det < 0)
31         det += 26;
32     detInv = modInverse(det);
33     if(detInv == -1)
34     {
35         printf("Plaintext matrix is not invertible.\n");
36         return 0;
37     }
38 }
```

Compiler Output:

```
- Warnings: 0
- Output Filename: C:\Users\sailp\Downloads\Crypto\Untitled1.exe
- Output Size: 130.4755859375 KiB
- Compilation Time: 0.11s
```



The screenshot shows the Dev-C++ IDE with the file 'Untitled1.cpp' open, displaying the continuation of the code. It calculates the inverse of the ciphertext matrix and then uses it to recover the key matrix. The key matrix is printed and then used to decrypt the ciphertext matrix to recover the plaintext matrix.

```
19     scanf("%d", &P[i][j]);
20     printf("Enter Ciphertext Matrix (2x2):\n");
21     for(i = 0; i < 2; i++)
22     {
23         for(j = 0; j < 2; j++)
24             scanf("%d", &C[i][j]);
25     }
26     det = (P[0][0] * P[1][1] - P[0][1] * P[1][0]) % 26;
27     if(det < 0)
28         det += 26;
29     detInv = modInverse(det);
30     if(detInv == -1)
31     {
32         printf("Plaintext matrix is not invertible.\n");
33         return 0;
34     }
35     Pinv[0][0] = (P[1][1] * detInv) % 26;
36     Pinv[0][1] = ((-P[0][1] + 26) * detInv) % 26;
37     Pinv[1][0] = ((-P[0][0] + 26) * detInv) % 26;
38     Pinv[1][1] = (P[1][0] * detInv) % 26;
39     printf("\nRecovered Key Matrix:\n");
40     for(i = 0; i < 2; i++)
41     {
42         for(j = 0; j < 2; j++)
43             K[i][j] = (C[i][0] * Pinv[0][j] + C[i][1] * Pinv[1][j]) % 26;
44         printf("%3d", K[i][j]);
45     }
46     printf("\n");
47     return 0;
48 }
```

Compiler Output:

```
- Warnings: 0
- Output Filename: C:\Users\sailp\Downloads\Crypto\Untitled1.exe
- Output Size: 130.4755859375 KiB
- Compilation Time: 0.11s
```

Output:

```
C:\Users\sailp\Downloads\Crypto\Untitled1.exe
Enter Plaintext Matrix (2x2):
7 8 9 3
Enter Ciphertext Matrix (2x2):
5 3 7 1

Recovered Key Matrix:
14 7
12 3

-----
Process exited after 13.24 seconds with return value 0
Press any key to continue . . .
```